

## Supplementary Information (Online Resource 6)

**Article title:** How Teams Decide Human Oversight in AI-Powered Systems: A Mixed-Methods Study

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## MCDA example

A compelling example that validates the integrated use of a structured questionnaire and Multi-Criteria Decision Analysis (MCDA), can be found in the domain of autonomous vehicles. Dubljević et al. [Dubljević] developed and applied a Multi-Attribute Impact Assessment (MAIA) questionnaire to systematically capture expert opinions on 13 potential harms and 8 potential benefits arising from different Automatic Vehicle (AV) deployment scenarios. This questionnaire served as the foundation for a formal MCDA process, enabling a structured comparison of competing system requirements such as safety, accountability, efficiency, and public trust. Their findings demonstrated that regulated AV deployment models that incorporate human oversight (e.g., fleet ownership) were ethically and socially preferable to fully autonomous or unregulated alternatives.

Once contradictions are identified, negotiation tools such as MCDA and Win-win method are used to quantify trade-offs between competing requirements. This structured decision-making approach allows stakeholders to assign relative importance to key AI system attributes based on business goals, regulatory constraints, and ethical imperatives.

To demonstrate how MCDA can guide the choice of human-oversight levels, we use the NTSB Highway Accident Report HAR-19/03 [NTSB] on the March 18, 2018 Tempe fatality, when an Uber ATG vehicle struck a pedestrian, as a work example. The pre-crash oversight setup had a single safety operator without real-time attentiveness monitoring, Volvo FCW (Forward Collision Warning)/AEB (Automatic Emergency Braking) disabled, and no emergency braking for impact mitigation. We evaluate three oversight alternatives: (1) Pre-crash baseline; (2) Post-crash robust - two operators, attentiveness monitoring, FCW/AEB enabled, mitigation braking permitted, and jaywalking modeled; and (3) Hybrid lean - a single operator with attentiveness monitoring while retaining the technical safety layers (FCW/AEB on, mitigation braking, jaywalking modeled). Alternatives are scored against five criteria: (i) expected crash-risk reduction, (ii) time-to-hazard response, (iii) operator workload/complacency, (iv) testing efficiency/throughput, and (v) cost. Weights emphasize safety (40%), then response time (20%), workload (20%), efficiency (10%), and cost (10%). The selection of criteria was informed both from the UBER incident, and by the author's professional experience in defense and transportation environments, where human oversight decisions directly influence system risk and mission outcomes. These criteria were further refined based on insights gathered from stakeholder interviews in the autonomous-vehicle industry, which highlighted the centrality of safety margins, intervention latency, and operator cognitive demands in real-time oversight scenarios.

The following table shows the results:

| Configuration  | Safety risk | Time to hazard | Workload | Efficiency | Cost | Weighted total |
|----------------|-------------|----------------|----------|------------|------|----------------|
| A. Pre-crash   | 1           | 1              | 1        | 5          | 5    | 1.8            |
| B. Post crash  | 5           | 5              | 4        | 3          | 2    | 4.3            |
| C. Hybrid lean | 4           | 5              | 3        | 4          | 3    | 3.9            |

*Table 1 – MCDA results*

Configuration A scores lowest because redundancy was removed (FCW/AEB off), braking was delayed/not allowed for mitigation, and a single operator increased complacency risk. B scores highest on safety and response by restoring FCW/AEB, allowing mitigation braking, modeling jaywalking, and adding two operators plus attentiveness monitoring. C is close to B, retaining the key safety layers but accepting single-operator workload risk.

The ATG Tempe case underscores the MCDA importance in RE for AI-powered systems, the crash emerged from interacting choices that reflected unexamined trade-offs among safety, efficiency, operator workload, and cost. Post-incident fixes improved safety but introduced new tensions, higher staffing and tooling costs, reduced testing throughput and added operator cognitive load. MCDA lets RE teams formalize these tensions early, define oversight alternatives, score them on criteria derived from hazards (risk reduction, response latency, workload, efficiency, cost, transparency), weight those criteria with stakeholders, and run sensitivity checks. This yields a traceable, evidence-based oversight requirement rather than an ad-hoc policy, and it documents why that level is justified given organizational priorities.